DSC–AC research series

Weekly correlations

Fourteen market variables, 2003–2026

Data validation, rolling estimates, and a Bayesian sampler run

Return sample	2003-08-15 to 2026-09-04
Weekly observations	1,204
Series / distinct pairs	14 / 91
Currency basis	Native quotes as supplied
Estimator status	Descriptive results plus a saved sampler run

Reading guide. The rolling correlations in this report are empirical estimates. The saved MATLAB run supplies the computational measurements. Unless the reported diagnostics establish otherwise, its draws remain preliminary. Full-sample smoothing and empirical-Bayes calibration use future information relative to an individual historical week.

September 2026

Executive summary

The weekly panel contains **1,204 dates, 14 variables, and 91 pairs**. Each weekly row is a Friday row: for each series, we use its last published level on or before that Friday. This keeps a local holiday in one market from deleting the whole week for every market. Japan's NKYTR series has one unusual 2019 holiday week; the affected returns are flagged so they can be checked separately.

The supplied sampler run completed 908 iterations in 720.0 minutes and retained 828 posterior draws. Its measured mean time was 189.54 seconds per completed iteration.

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1 Data and weekly alignment

The raw file contains levels, rather than daily returns. The weekly panel uses each series' last available observation on or before the Friday endpoint.

$$L_{jt} = P_j(s_{jt}), \qquad s_{jt} = \max\{s \le f_t : P_j(s) \text{ is observed}\}, \qquad y_{jt} = 100\,[\log L_{jt} - \log L_{j,t-1}]\,. \quad (1)$$

Here f_t is the Friday date. Endpoint age means how old the value is when placed in that Friday row: a Friday close has age 0, a Thursday close has age 1, and a Tuesday close has age 3. Routine cases may use a value up to four calendar days old. The single exception is NKYTR on 3 May 2019: Japanese markets were closed for the extended holiday around the imperial succession, so the Friday row uses the previous published level from 26 April, seven days earlier (Japan Exchange Group, 2019).

The source starts on Monday 4 August 2003, so the first completed Friday level is 8 August 2003. A return needs two weekly levels; therefore the first return is from 8 August to 15 August 2003. The week ending 11 September 2026 is incomplete and excluded. The result has 1,205 level dates and 1,204 return dates.

2 Descriptive return results

All analyses below use percentage weekly log returns. Standard deviations are weekly, with no annualization. Series names follow the supplied labels, avoiding unverified instrument descriptions.

Ticker	N	Mean (%)	SD (%)	1st pct.	99th pct.
AUDUSD	1204	0.008	1.695	-4.35	4.27
BCOMCOT	1204	0.127	4.666	-12.14	10.56
BCOMGCTR	1204	0.193	2.450	-7.12	6.18
EURUSD	1204	0.002	1.232	-3.37	2.97
GBPUSD	1204	-0.014	1.295	-3.69	2.82
I02981JP	1204	0.007	0.338	-0.94	0.85
LEATTREU	1204	0.049	0.629	-1.74	1.73
LSG1TRGU	1204	0.052	1.025	-2.72	2.53
LUATTRUU	1204	0.052	0.617	-1.51	1.58
NKYTR	1204	0.194	2.969	-7.28	6.91
SPXT	1204	0.208	2.380	-6.74	6.00
SX5T	1204	0.131	2.821	-7.78	6.50
TUKXG	1204	0.151	2.278	-6.74	5.92
USDJPY	1204	0.023	1.350	-3.59	3.28

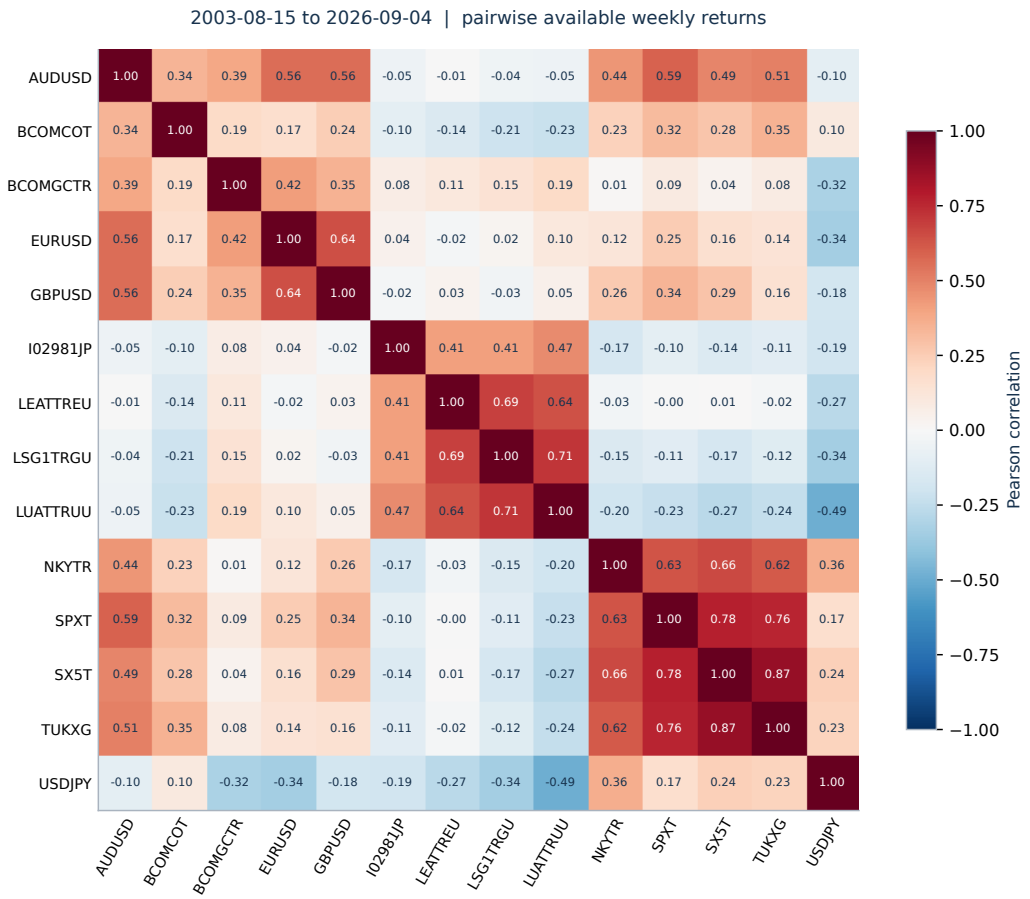


Figure 1: Full-sample Pearson correlations, 2003-08-15–2026-09-04. The baseline retains flagged as-of marks around the closure. Source: supplied market levels and generated weekly-research outputs. Native quotes; descriptive estimates.

3 Stationarity of weekly returns

We test the 14 weekly return series $y_{jt} = 100\Delta \log L_{jt}$. Table 1 reports augmented Dickey–Fuller (ADF) tests with a constant and BIC-selected lags from 0–13 (MathWorks, 2026). ADF rejects a unit root for 14 of 14 return series at the 5% level. The largest ADF p-value shown in the table is 0.001 (subject to the tabulated bounds in the table).

Table 1: ADF unit-root tests for weekly returns.

Ticker	ADF lag	ADF statistic	p-value
AUDUSD	0	-36.245	≤ 0.001
BCOMCOT	0	-33.002	≤ 0.001
BCOMGCTR	0	-35.649	≤ 0.001
EURUSD	0	-34.942	≤ 0.001
GBPUSD	1	-26.761	≤ 0.001
I02981JP	0	-36.053	≤ 0.001
LEATTREU	0	-34.087	≤ 0.001
LSG1TRGU	0	-35.935	≤ 0.001
LUATTRUU	0	-36.459	≤ 0.001
NKYTR	0	-35.211	≤ 0.001
SPXT	0	-37.097	≤ 0.001
SX5T	0	-37.028	≤ 0.001
TUKXG	0	-37.327	≤ 0.001
USDJPY	0	-36.451	≤ 0.001

Null: a unit root. Lag selection uses a common 1,190-observation regression sample; the full return panel remains available for estimation. The p-value bounds preserve MATLAB's tabulated limits. Source: saved `stationarity.csv`; method: MathWorks (2026).

Conclusion. The evidence supports treating the weekly returns as stationary for this analysis, without further differencing.

4 Joint Bayesian correlation model

The goal of the Bayesian model is to estimate a time series of correlations after accounting for each market's own mean and volatility. In each week, the model separates the return vector into three moving parts: expected returns, individual risk levels, and the correlation matrix linking the return innovations. The object of interest is the sequence of correlation matrices P_t, not the raw return correlations and not the sampler's internal coordinates.

4.1 Observation and state equations

The repository implements the dynamic stochastic-correlation approach associated with Arias et al. (2023). For $m = 14$ and $q = m(m-1)/2 = 91$, retain its time-varying intercept specification, with no VAR lags ($p = 0$):

$$y_t = B_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, H_t), \tag{2}$$

$$H_t = D_t P_t D_t, \qquad D_t = \mathrm{diag}\{\exp(h_{1t}/2), \ldots, \exp(h_{mt}/2)\}, \tag{3}$$

$$B_t = B_{t-1} + \eta_t, \quad t \geq 2, \qquad \eta_t \sim \mathcal{N}(0, V), \tag{4}$$

$$h_{jt} = h_{j,t-1} + u_{jt}, \qquad u_{jt} \sim \mathcal{N}(0, \sigma^2_{h,j}), \tag{5}$$

$$r_{kt} = r_{k,t-1} + v_{kt}, \qquad v_{kt} \sim \mathcal{N}(0, \sigma^2_{r,k}). \tag{6}$$

The unrestricted vector $r_t = \mathrm{vecl}(\log P_t)$ is the lower off-diagonal part of the matrix logarithm of the correlation matrix. Reconstruction follows Archakov and Hansen (2021). These coordinates are not correlations. Numerical outputs therefore store actual distinct entries of P_t, alongside the latent states needed for checkpointing. There are 119 time-varying scalar states per week: 14 means, 14 log variances, and 91 correlation coordinates.

In the baseline dataset used here, all 14 return series are present for every one of the 1,204 weeks, so the model uses the full 14-dimensional return vector each week:

$$y_t \mid B_t, h_t, r_t \sim \mathcal{N}\left(B_t, H_t\right). \tag{7}$$

The Kalman filter and backward simulation smoother use this full weekly vector. Historical smoothing conditions on the full sample; a smoothed estimate at time t is not a real-time estimate available at time t.

4.2 Prior definitions

$$B_1 \sim \mathcal{N}(\bar{B}, 4\bar{V}_B), \qquad V \sim \mathrm{IW}(V_{0B}, \nu_B), \qquad B_1 \perp V, \tag{8}$$

$$\sigma^2_{h,j} \overset{\text{ind}}{\sim} \mathrm{IG}(a_h, b_h), \qquad h_{j0} \mid \sigma^2_{h,j} \sim \mathcal{N}(\bar{h}_j, 10\sigma^2_{h,j}), \quad j = 1, \ldots, m, \tag{9}$$

$$\sigma^2_{r,k} \overset{\text{ind}}{\sim} \mathrm{IG}(a_r, b_r), \qquad r_{k0} \mid \sigma^2_{r,k} \sim \mathcal{N}(\bar{r}_k, 10\sigma^2_{r,k}), \quad k = 1, \ldots, q. \tag{10}$$

Here $m = 14$, $q = 91$, and h_{j0}, r_{k0} are pre-sample states. Prior blocks are independent; initial volatility and correlation states are conditionally independent given their innovation variances. Hyperparameter definitions and empirical-Bayes calibration are recorded in the saved run artifacts.

5 Sampler run

This section records the saved sampler run used for the current report. The run stopped at the configured time cap, so convergence is reported from the saved diagnostics rather than inferred from completion.

Run quantity	Value
Initial-prior weeks excluded	104
Weekly returns in the likelihood	1100
Model sample	2005-08-12 to 2026-09-04
Independent chains	4
Completed iterations (sum across chains)	908
Retained posterior draws (pooled)	828
Elapsed wall time	720.01 minutes
Mean time per completed iteration	189.54 seconds
Stop status	Time cap reached
Warm-up iterations per chain	20

6 Run diagnostics

This section reports the saved convergence diagnostics for the sampler run. The reference values are diagnostic thresholds for judging whether independently initialized chains agree well enough for the reported posterior summaries.

Diagnostic	Reference value	Observed	Result
Maximum rank-normalized split $\widehat{R}$	< 1.01	2.715	Fail
Minimum bulk ESS	≥ 400	4.853	Fail
Minimum tail ESS	≥ 400	11.03	Fail
Maximum MCSE/SD	≤ 0.05	0.4712	Fail

Overall diagnostic status: failed. Retained draws by chain: 209, 206, 204, 209. Aligned diagnostic draws per chain: 204; checked quantities: 130,900. Unequal retained counts: aligned diagnostics are exploratory because pooled summaries use all retained draws. Convergence is not established.

References

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